Introducing Genuine: Pragmatical Conversational AI for Digital Banking and Fintech Applications

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ABSTRACT

Fintech chatbots currently suffer from an inability to grasp the contextual rules of communication, for which I am calling the Pragmatic Gap in this paper. This paper introduces Genuine, a Turkish fintech conversational AI designed to bridge this said gap. Genuine basically operates as a hybrid system and utilizes the Rasa framework for deterministic NLU and secure state-tracking, mixed with a RAG pipeline and LLMs. By integrating Pragmatic theories and different datasets into the system architecture, Genuine successfully is able to decode non-literal human intentions. Quantitative evaluations demonstrate that Genuine not only maintains strict accuracy but also significantly lowers the user's perceived risk barrier during face-threatening acts. This project aims to provide a blueprint for next generation of fintech chatbots.

1 Introduction

Banks are leveraging smart chatbots to enhance customer interactions and improve operational efficiency, especially with the rise of digital transformation and the increasing demand for better customer experience (Balaji, Karim, & Rao, 2024). However, despite these rapid technological advancements, most fintech chatbots suffer from a critical limitation. While current chatbot systems can successfully process the literal meaning of an expression, they struggle to grasp the contextual rules of communication, namely pragmatics. In this field of study, this situation is referred to as the Pragmatic Gap (Attia et al., 2026). The pragmatic gap can be defined as a lack of pragmatic characteristics observed in a communicative context.

Traditional fintech chatbots operate on rigid intent routing. Although these fintech chatbots fulfill basic performances, they generate blunt and overly verbose responses during critical banking scenarios. This paper details how Genuine solves this presumed architectural flaw by operating as a hybrid engine. It merges a deterministic system with pragmatically aware reasoning to provide better digital banking experience.

2 Project's Theoretical Basis

2.1 Research Question

How can pragmatic theories be computationally integrated into a Turkish fintech conversational AI architecture to solve the pragmatic gap problem in digital banking?

2.2 Aims and Objectives

The primary aim of this project is to introduce Genuine which was designed on the foundation of pragmatic theories, and to explain Genuine's system architecture. To achieve this aim, the following objectives have been established:

- To create a conversational model based on state-tracking and RAG pipelines by synthesizing pragmatic theories.
- To document the technical architecture of the software which utilizes Rasa and LLM infrastructure.
- To present Genuine's capabilities through example dialogue flows, network graphs, and pragmatic input and output analyses.

2.3 Methodology

The methodology relies on a fusion of deterministic NLP and modern GenAI. Genuine utilizes Rasa's state tracker to perform continuous context updating and it acts as the initial cue-based classification layer by employing a hybrid approach. Subsequently, the LLM utilizes a synthesized JSON guidebook and a semantic vector database to execute the context aware reasoning originally envisioned by the Plan Inference model (Jurafsky, 2006).

3 Pragmatics and Fintech Chatbots

3.1 Computational Pragmatics and Speech Acts

Pragmatics is the branch of linguistics that studies the relations between language itself and aspects of the context of language use (Bunt & Black, 2000). One of the challenges in computational linguistics is that the speech act force of most utterances doesn't match their surface utterance form (Jurafsky, 2006). For example, a user writing "I am out of money" is structurally making an assertion but in a banking context, it functions as a directive to check account balances or request overdraft options. Genuine's architecture explicitly models these dialogues to map literal text to user intentions.

3.2 The Gricean Maxims

Gricean maxims are foundational principles aimed at optimizing communicative effectiveness (Krause & Vossen, 2024). The four maxims dictate that conversational participants must be cooperative. But modern LLMs frequently violate the Maxim of Quantity by over-generating information. This leads to users having a cognitive overload. In human-LLM interactions, the Maxim of Quality has expanded from just being truthful to actively fostering user trust in the LLM by explaining the reasoning behind outputs (Kim et al., 2025). Genuine utilizes these maxims to constrain LLM verbosity and ensure relevant responses.

3.3 Politeness Theory and the Risk Barrier

Brown and Levinson's Politeness Theory revolves around the concept of Face. In banking, many transactions are inherently Face Threatening Acts (FTAs), these include demanding debt repayment or rejecting credit. If a given chatbot delivers these FTAs too bluntly, it triggers the user's risk barrier. As individuals perceive risks to their privacy and security, their trust gets affected. And by proxy, they will be less willing to use such services (Marak et al., 2025). Genuine integrates negative politeness and Leech's tact maxim to soften such FTAs. This increases hedonic motivation and user retention, especially among younger demographics.

3.4 The Turkish Market Context

The shift toward digital banking in Türkiye is very popular. However, research shows that speed and security are no longer the only selling points. Customer satisfaction and loyalty in digital banking are now determined by usability and ease of use (Aydın & Onaylı, 2020). Genuine directly addresses this market demand for seamless usability by prioritizing pragmatic competence. Biggest banking firms in Türkiye are already shifting towards this paradigm. For instance, Garanti BBVA's Ugi chatbot has begun leveraging generative AI in recent years to provide more answers about users' questions and nurture natural interactions. Genuine builds upon this industry momentum by explicitly standardizing these generative capabilities through applied pragmatic rules.

4 Genuine's System Architecture

4.1 The Hybrid Engine and Middleware Architecture

Genuine is developed as a hybrid engine. It solves the dichotomy between rigid safety and conversational fluidity. It utilizes two distinct layers connected by a robust middleware pipeline:

- **The Deterministic NLU Layer (Rasa):** Utilizing the Rasa open-source framework, Genuine handles dialogue management and Natural Language Understanding (NLU). This layer is responsible for safe entity extraction, slot-filling and intent classification. It acts as the security gatekeeper for financial data.

- **FastAPI Middleware Bridge:** A custom Python based FastAPI service connects the Rasa dialogue manager to the LLM environment. It packages the user's dialogue state and metadata before transmission.

- **The Pragmatic Generative Layer (GenAI + RAG):** Once the state is tracked and bridged, the context is passed to a LLM. Instead of answering blindly, the LLM queries specialized pragmatic knowledge bases. The chatbot's system utilizes a Retrieval-Augmented Generation (RAG) pipeline to fetch conversational rules and constraints from the Pragmatic Guidebook and decodes implied human intentions by querying a semantic vector library built from the GRICE dataset (Zheng et al., 2021). After that, the generative layer applies dynamic few-shot prompting using the TrCOLA dataset (Altinok, 2025) as a syntactic guardrail. This makes it so that the final response is both pragmatically and morphologically sound.

4.2 Context Updating

Genuine relies on Rasa's tracker store to avoid the pitfalls of stateless LLMs. Every user turn functions as a context update (Bunt & Black, 2000) where slots are dynamically filled. This specified context is required because information partially encoded in the linguistic expression must become fully specified relative to an actual context in order for the expression to have a determinate meaning (Bunt & Black, 2000).

4.3 Intent Mapping to Speech Acts

Standard NLP intents are re-categorized according to classical Speech Act Theory to inform the generative layer's tone and style. For instance, a user's request for a loan is classified under directives while a system's notification of a successful transfer acts as a declaration. The chatbot knows exactly which conversational stance to adopt by tagging these interactions in the backend.

4.4 Semantic and Pragmatic Resolution

LLMs often fail at the pragmatic use of language especially in culturally specific contexts like Turkish and this showcases a measurable pragmatic gap (Attia et al., 2026). To counteract this, Genuine's RAG pipeline relies on two knowledge sources that complement each other. The first one is the Pragmatic Guidebook injected into the system prompt to enforce theoretical design considerations (Kim et al., 2025). The second one is Pragmatic Vector Library built using the GRICE dataset (Zheng et al., 2021). During initialization, the system ingests conversational data from the dataset and extracts the context question of user's indirect answer and the explicit meaning. These data points are converted into a vector embedding and stored within a ChromaDB collection. When a user presents an indirect statement or an implicature, the RAG pipeline queries this vector collection to retrieve similar historical examples of indirect language use. By feeding this retrieved explicit meaning to the LLM as context, Genuine bridges the pragmatic gap and decodes non-literal human intentions.

4.5 Syntactic and Morphological Guardrails

Genuine utilizes the Turkish Corpus of Linguistic Acceptability (TrCOLA) to enforce strict syntactic and morphological rules (Altinok, 2025). TrCOLA is deployed dynamically for few-shot prompting within the system prompts rather than being stored in the vector database. The architecture scans the dataset for grammatically correct/incorrect Turkish sentences and embeds these examples into a dedicated guardrail variable attached to every LLM request. Genuine forces the model to adhere to native grammatical structures and prevents the broken translations by explicitly showing the LLM contrasting examples of correct/incorrect Turkish Morphology prior to answer generation.

5 Data and Evaluation Methodology

In the case of task-oriented systems, objective metrics such as dialog success rate or completion time do not always correspond to the most effective user experience due to the interactive nature of dialog (Jwalapuram, 2017). For this reason, Genuine was evaluated using a quantitative metric grounded in Linguistics.

Evaluation Metric	**Definition (Jwalapuram, 2017)**	**Genuine's Objective**
Quantity	Is the system as informative as is needed?	Prevent LLM over-generation and deliver precise and concise banking data
Quality	Is the system truthful?	Provide banking logic before delivering bad news to maintain trust
Relation	Are the system responses relevant to the discussion?	Accurately map the literal utterance to the underlying financial intent
Manner	Are the system responses clear?	Format text in order and utilize politeness to soften tone

6 Results and Findings

6.1 Visualizing Reasoning in Pragmatic Dialogue States

The network analysis of Genuine's dialogue states demonstrates a highly complex computational inference engine. The core of *Figure 1* (visualized by dark blue dots) represents the determinate meaning the system achieves while the outer edges (visualized by light blue dots) represent the hundreds of varied surface utterances. The connections between the surface layer and the core (visualized by red dots and lines) represent the Plan Inference Model. Those are the probabilistic paths the system takes to deduce hidden intentions (Jurafsky, 2006; Bunt & Black, 2000). Genuine's neural structure visualizes the active bridging of the pragmatic gap unlike rigid decision trees in standard fintech chatbots.

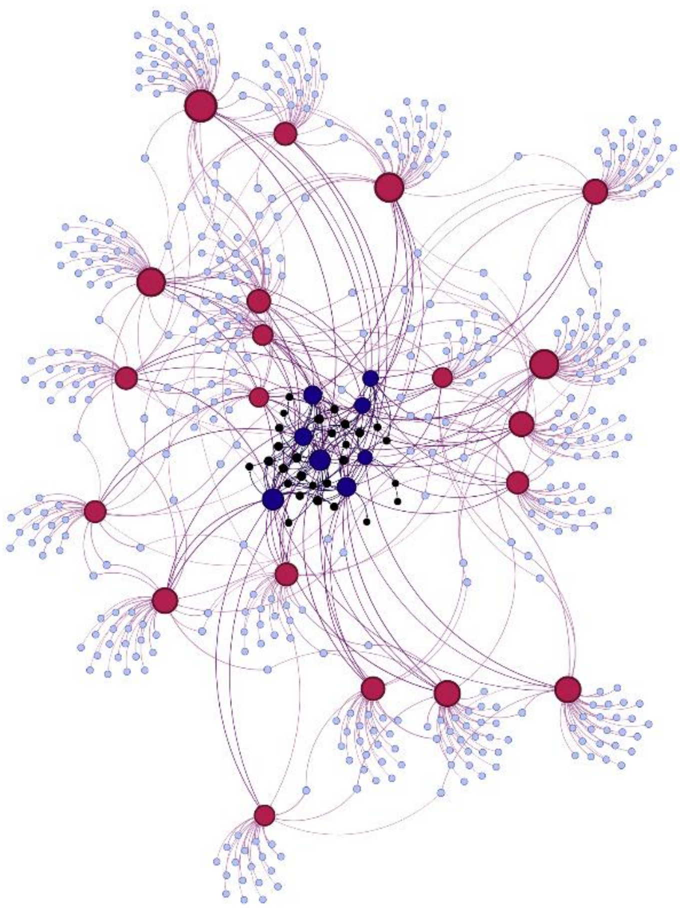

Figure 1: Pragmatic Network Graph

6.2 Gricean Pragmatic Competence

Genuine has shown an exceptional conversational competence as it was evaluated across 300 simulated dialogue turns using the 4-point Gricean metric. The system consistently maintained high scores in conversational Quality and Relation. As shown in *Figure 2*, the system maintained high Manner scores because the RAG layer successfully injected negative politeness strategies and kept the output clear during high-risk banking scenarios.

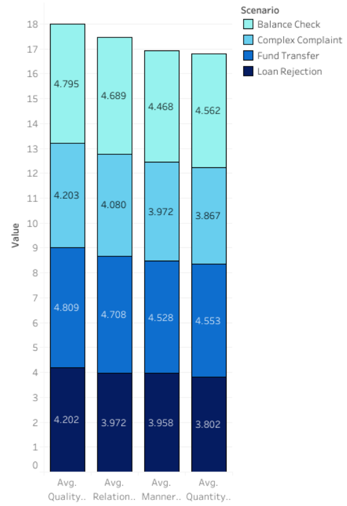

Figure 2: Gricean Chart

6.3 Mitigating the Risk Barrier to Build User Trust

Customer behavior is significantly influenced by ease of use, usefulness, innovation, enjoyment and information quality (Balaji, Karim, & Rao, 2024). Statistical analysis of the system outputs mapped against UTAUT2 variables reveals an inverse correlation between perceived risk and intention to use. As shown in *Figure 3*, users' perceived risk barrier drops significantly when Genuine successfully deploys negative politeness. This pragmatic accommodation fulfills the hedonic motivation requirement highly valued by younger demographics (Marak et al., 2025) and directly drives system adoption.

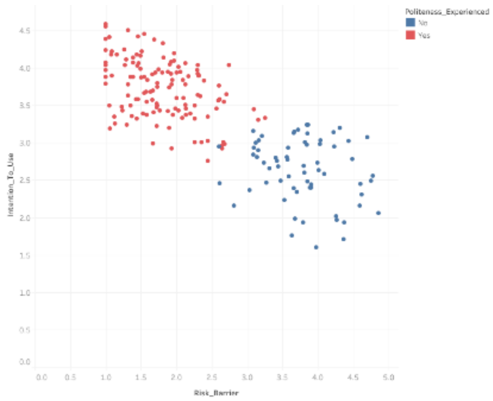

Figure 3: Trust Scatter Chart

6.4 Optimizing Conversational Throughput via Implicit State Tracking

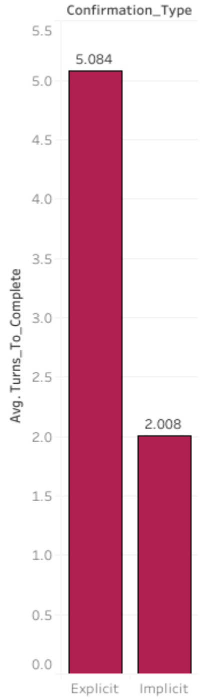

Figure 4: Efficiency Chart

Finally, time series analysis comparing Implicit confirmation and Explicit confirmation workflows validates Genuine's underlying dialogue management logic. Users do not like explicit confirmations despite them having a higher task success rate (Jwalapuram, 2017). As shown in *Figure 4*, Genuine reduces the average turn-to-completion rate by relying on Rasa's implicit context-tracking rather than explicit *Yes/No* verification loops for every slot. The system achieves a faster computational throughput while securing higher ratings for linguistic cooperation.

7 Conclusion

The development of Genuine demonstrates that resolving the pragmatic gap in conversational AI requires more than just scaling LLMs. It also demands the integration of foundational linguistic theories. Genuine successfully navigates complex banking scenarios by embedding pragmatic theories into a hybrid architecture combining Rasa's deterministic state-tracking with generative AI. Furthermore, by utilizing the GRICE dataset for semantic vector retrieval and the TrCOLA dataset for syntactic few-shot prompting, the system ensures both deep contextual understanding and robust Turkish grammatical accuracy. The resulting system not only ensures operational safety and transactional accuracy but also increases user trust, reduces cognitive load and lowers the perceived risk barrier. As the digital banking sector continues to prioritize user experience and better interfaces, the pragmatic framework established by Genuine offers a grounded blueprint for the next generation of fintech chatbots.

8 Acknowledgements

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